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Vinay Ram Gazula

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I am a passionate Mechanical Engineer and Data Science enthusiast eager to advance my career as a Data Scientist/Analyst. I am a well-organized and highly driven guy who always performs to the best of his capabilities. I contribute to open-source projects and I am open to work on projects involving Data Science, AI/ML, FinTech and CFD.

Skills

Programming	Python*, R*, SQL, Bash*, C/C++, Matlab	
Data Science	Spreadsheets, Pandas, Tidyverse, Matplotlib, Plotly, Tableau	
Tools	**Git**	GitHub*
	IDE's	KDE Advanced Text Editor (KATE)*, VS Code, VIM
	Documentation	MS Office, LaTeX*, Markdown*
	Operating Systems	Windows, Linux*
Libraries	NumPy, SymPy, yfinance, Tensorflow, ggplot	
CAD & CFD Softwares	Solidworks*, AutoCAD*, CATIA, Tecplot 360*, Gmsh*, ANSYS Fluent	

* Frequently used

Projects

MURPH Analytics — **Mar 2023 - Ongoing**
Personal Project — *Project Source*

- Built an interactive dashboard using Streamlit to post-process and perform different analytical studies on the simulation data generated from the MURPH Solver.
- Implemented unstructured mesh plotting and 2D contour plots using Plotly.

MURPH — **Dec 2022 - Ongoing**
Personal Project — *Project Source*

- A Multi-dimensional (2-D/3-D), Unstructured-mesh and Reactions based Parallel solver for Hyper-sonic flow regimes.
- New design uproach is used to build the solver code so that the parallel code generation can be fully automated.
- 2D Ideal gas solver code is developed and it is in validation phase.

AlgoTrade API — **May 2022 - Oct 2022**
Personal Project — *Project Source*

- Fully automated NSE Stock/Equity trading-bot for Kotak Securities (KOTAK TRADE API) with integrated back testing and ML algorithms for price and trend predictions.
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API.

Accelerating a Hypersonic CO_2 Reaction Solver — **Jul 2020 - May 2021**
B.Tech Final Year Project — *Academic Project*

- A 2-D unstructured finite volume method based solver capable of simulating hypersonic flows in CO_2 rich Martian atmosphere with chemical kinetics for eight species consisting of $N_2, O_2, NO, O, N, CO_2, CO$ and C incorporated in it is developed using python.
- The solver is parallelized to run on single or multiple CPUs and Nvidia GPUs in order to attain greater computational speeds using OP2 application programming interface.

Design and Analysis of Heat Sink — **Oct 2020 - Dec 2020**
Thermal Design of Electronic Equipment Course — *Coursework Project*

- Designed and analysed a 2-D heat sink made of Aluminium and a layer of epoxy between the chip and heat sink with specified thermal properties as a thermal design solution for cooling a noval chip (CPU for example) using CFD-HT (Computational Fluid Dynamics and Heat Transfer) software ANSYS Fluent.

Professional Certifications

Google Data Analytics — **Dec 2022 - Apr 2023**
Coursera — *Certificate*

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Publications

Accelerating a Hypersonic CO_2 Reaction Solver **Dec 2021**
ISHMT-ASTFE Heat and Mass Transfer Conference *Publications*

- Vinay, G., Dinesh, D., Satya, P., "Accelerating a Hypersonic CO_2 Reaction Solver", ***Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference December 17-20, 2021, IIT Madras, Chennai-600036, Tamil Nadu, India.***

Experience

Research Intern **Sep 2021 - Feb 2022**
Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) *SRM University AP*

- Worked on a Science Education and Research Board (SERB) govt. of India project titled "Wall Effects in Shock Wave Boundary Layer Interaction".
- Contibuted in the development of solver code for gaseous reactions and viscous properties.
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc,.
- Published a conference paper in International ISHMT-ASTFE Heat and Mass Transfer Conference.

Internship on CATIA **Jun 2020 - Aug 2020**
AP State Skill Development Corporation (APSSDC) *Remote*

- Practical experince with Catia V5 software.
- Designed a Turbocharger assembly for automobiles using Catia.

Education

Bachelor of Technology in Mechanical Engineering **2017-2021**
SRM University, AP *CGPA: 8.51*
Awarded with merit scholarship of 50% tuition fee.

Board of Intermediate Education Andhra Pradesh, BIEAP (12th) **2015-2017**
Narayana Jr. College *Percentage: 95.1%*

Central Board of Secondary Education, CBSE (10th) **2014-2015**
Sri Chaitanya Techno School *CGPA: 9.4*

Certifications

Algorithmic Trading & Quantitative Analysis Using Python **Mar 2023 - Apr 2023**
Udemy *Certificate*
Introduction to Programming with MATLAB **Apr 2020 - May 2020**
Coursera *Certificate*
Autodesk Fusion 360 Integrated CAD/CAM/CAE **Feb 2020 - Mar 2020**
Coursera *Certificate*
Robo Feast 2018 **Feb 2018**
Certificate of Participation
Participated in the 2 days National Level Workshop cum Championship on "*WiFi-Controlled Robotics*" organized by ***Revert Technology Pvt. Ltd. & EDC, IIT Roorkee.***